

High-Rise Society Management Application

Complete Technical Requirements & Functional Specifications Document

BUILDING PROFILE

22 Floors | 154 Units

DEPLOYMENT COST TARGET PLATFORM TYPE

\$0.00 / Month (100% Free)

Responsive Web App (PWA)

TARGET PLATFORM

Antigravity Engine

1. Project Scope & Operational Overview

This document specifies the operational logic, user workflows, data schema specifications, and zero-cost infrastructure architecture for a full-fledged society web application designed for a 22-floor residential tower housing 154 families.

Primary Objective: Eliminate resident friction, provide 100% financial transparency, streamline common area maintenance, and handle building emergency communications without recurring infrastructure costs.

2. User Roles & Permission Matrix

Role Type	Target Users	Primary Permissions & System Access
Resident User	Flat owners, co-residents, tenants (154 units)	Raise public tickets, view live financial ledger, upload payment receipts, book amenity slots, receive broadcast alerts.
Management Admin	Society Secretary, Treasurer, Committee Members	Approve/verify payment slips, record society expenses, update ticket resolution status, post pinned notices, manage amenity blocks.
Auditor / Public View	All Building Members (Transparent View)	Read-only view of society expenses, balance breakdowns, maintenance payment status per flat, resolved ticket logs.

3. Functional Specifications (Core Modules)

3.1 Transparent Ticket & Maintenance Board

1. Issue Creation

Resident submits title, category (Elevator, Plumbing, Lighting, Cleanliness, Parking, Security), detailed text description, location tag (e.g., *Floor 14 Common Hallway*), and attaches compressed photo proof.

2. Real-Time Broadcast Alert

System automatically drops a real-time event notice to all 154 flats. Prevents duplicate ticket submission for public area issues.

3. Admin Assignment & Progression

Admin sets priority level (Urgent / Normal / Scheduled), assigns vendor/maintenance staff, and updates status (*Open* → *In Progress* → *Resolved*).

4. Resolution & Proof Notification

Once marked *Resolved*, the app notifies all residents with "After" photo proof attached. The original reporter has 24 hours to reopen or rate the service.

3.2 General Announcement Board

- **Categorized Notices:** Tagged as **EMERGENCY**, **MAINTENANCE**, **COMMUNITY**, or General Notice.
- **Pinning Engine:** Critical announcements stay fixed at the top of the resident dashboard until dismissed or unpinned by admins.
- **Read Audit Counter:** Tracks flat-level view receipts (e.g., "*Acknowledged by 128 / 154 Flats*") so management knows when notices are seen.

3.3 Community Expense & Fund Tracker

Payment Deposit Workflow:

1. Resident completes monthly maintenance payment via UPI to the society bank account.
2. Resident uploads transaction reference ID, payment month, paid amount, and screenshot of the UPI transaction receipt.
3. System marks entry status as **PENDING VERIFICATION**.
4. Admin verifies screenshot against bank receipts and clicks **Approve**. Entry transitions to **VERIFIED**.

Debit & Outflow Tracker:

- Admin registers all outgoing society funds (e.g., Cleaner wages, Security agency payment, Electricity bills, Elevator AMC).
- Admin attaches mandatory vendor receipt/invoice soft copy for public auditing.

Public Financial Dashboard: Live public formula computes total verified deposits minus total society expenses to display the current remaining net balance of the building fund.

3.4 Amenity Booking System

- **Supported Facilities:** Clubhouse, Terrace Garden, Party Lawn, Gym, Tennis Court.
- **Conflict Prevention Rules:** Automatic slot validation preventing overlapping booking requests for the same date/time window.
- **Fair Use Policies:** Maximum limits enforced per flat (e.g., maximum 1 weekend party booking per month per unit).

3.5 Emergency Broadcast & Contact Directory

- **One-Tap Push Alerts:** Direct browser push notifications to desktop and mobile devices for sudden water/power outages or fire alerts.
- **Telegram Bot Fallback Integration:** Simultaneous emergency alert dispatch to the building's main Telegram group via automated API trigger.
- **Offline Contact Caching:** Critical numbers (Security Desk, Elevator Technician, Electrician, Plumber, Nearby Emergency Hospitals) stored in local browser memory for offline access.

4. Database Architecture & Data Entities

The system requires 6 core relational data entities structured as follows:

Entity Name	Attributes / Fields	Relationships
Flats	ID, Flat Number, Floor Number (1-22), Owner Name, Resident Contact, Role (Resident/Admin).	One-to-Many with Tickets & Payments
Tickets	ID, Flat ID (Foreign Key), Title, Description, Category, Photo URLs, Status (Open/In Progress/Resolved), Timestamps.	Belongs to Flat
Payments	ID, Flat ID (Foreign Key), Amount, Month/Year, Transaction Ref, Screenshot URL, Status (Pending/Verified/Rejected).	Belongs to Flat
Society Expenses	ID, Title, Category, Amount Paid, Vendor Name, Invoice/Receipt URL, Date Paid, Approved By Admin ID.	Standalone Ledger Entry
Announcements	ID, Title, Content, Priority Category, Is Pinned (Boolean), Created At, Author Admin ID.	One-to-Many with Read Audit
Amenity Bookings	ID, Flat ID (Foreign Key), Amenity Name, Start Time, End Time, Status (Confirmed/Cancelled).	Belongs to Flat

5. Zero-Cost Infrastructure & Limits Strategy

Strategy for Staying \$0.00 / Month Forever:

- **Client-Side Image Compression:** All ticket photos and UPI payment screenshots are compressed directly in the resident's web browser down to ~150 KB prior to server upload. This extends 1 GB free cloud storage capacity to handle over 6,500 file uploads.
- **Progressive Web App (PWA):** App can be installed directly onto Android/iOS home screens via browser prompts without paying Apple or Google developer store fee subscriptions.
- **PostgreSQL Free Tier Optimization:** Auto-purge or archive resolved ticket logs and verified receipts older than 24 months to maintain database size well under the 500 MB free allocation limit.

6. Non-Functional Requirements

- **Mobile-First Responsiveness:** Optimized for quick phone browser usage by residents and guards.
- **Data Security & Isolation:** Row Level Security policies ensure residents can only edit their own profile and payment submissions, while financial totals and ticket boards remain transparently readable building-wide.
- **Uptime & Availability:** High availability serverless infrastructure ensuring 99.9% uptime during emergency broadcasts.